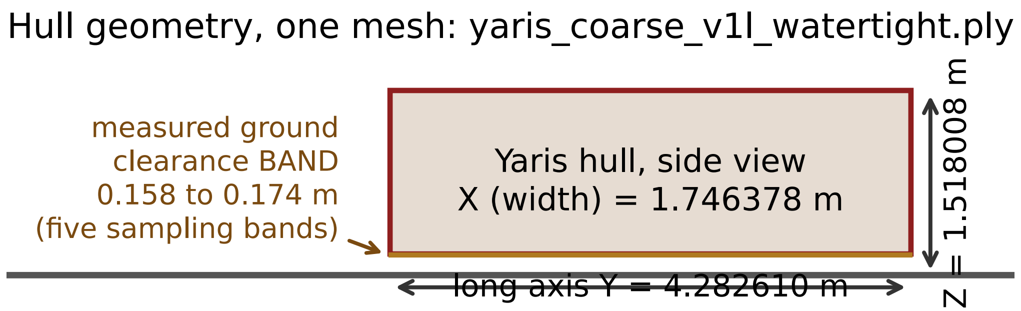
G7 Geometry pipeline and the gate table, including what fails



Geometry and physics gates, measured this session

gate	measured value	verdict
mesh containment of solid particles	100.00 pct of 2000 sampled	**PASS**
fill_ratio vs 0.95 to 1.10 band	1.0024 at g64; 0.9941 to 1.0262 (17 runs)	**PASS**
water surface open_edges	0	**PASS**
water volume error	+1.68% raw, +1.65% smoothed	**PASS**
oob_particle_frames	0	**PASS**
determinism_identical, all 17 runs	True	**PASS**
realized rho vs the 100 to 300 band	310 / 453 / 658 kg/m3 at grid 64	**FAIL**
water particle passthrough	7.3 to 15.9 pct; worst at v = 3.0 m/s	**NOTED**

The gate table is shown WITH its failure. A gate table with no failures is not credible, and the density gate genuinely fails: solidify_watertight fills the hull to parity (fill_ratio ~ 1.00), so the realized effective density is set by mass over a correctly measured 3.5427 m3 hull volume and lands at 310 / 453 / 658 kg/m3, above the project's own stated 100 to 300 kg/m3 plausibility band. The band is the thing that is now wrong for a parity-filled hull, but it is the project's stated band and the run is outside it, so it is recorded as FAIL rather than quietly rewritten.
Containment is 100.00 pct of a 2000-particle SUBSAMPLE, not of all particles: trimesh.ray.has_embree is False on this machine so the gate was run subsampled (docs/render_v1_task_outputs/A2_containment_gate.output). The four-rotation relative gate that would discriminate the pose chain is still outstanding.
Volume error and open_edges are read from the FIXED PyVista render logged this session in renders/yaris_render_s1/hero_fixed.log. Passthrough is a NOTED caveat, not a pass: 7.3 to 15.9 pct of water particles are inside the hull at the worst frame, worst at the highest surge velocity, which is a real limit on how hard the high-velocity end of G1 can be pushed.
Extents and clearance sources: rollout.npz extent field; docs/four_rung_ladder.md:274-280 for the 0.158 to 0.174 m measured clearance band.